# MAT2002 HW6 文杰 123090612

## 1

### (a)

For $p > 0$,

$$\Gamma(p+1) = \int_0^\infty e^{-x} x^p dx.$$

Use integration by parts:

$$u = x^p, \qquad dv = e^{-x} dx.$$

Then

$$du = p x^{p-1} dx, \qquad v = -e^{-x}.$$

Hence

$$\begin{aligned}\Gamma(p+1) &= \int_0^\infty e^{-x} x^p dx \\ &= \left[-e^{-x} x^p\right]_0^\infty + p \int_0^\infty e^{-x} x^{p-1} dx.\end{aligned}$$

Since $p > 0$,

$$\lim_{x\to\infty} e^{-x} x^p = 0, \qquad \lim_{x\to 0^+} x^p = 0.$$

So the boundary term is 0, and therefore

$$\Gamma(p+1) = p \int_0^\infty e^{-x} x^{p-1} dx = p\Gamma(p).$$

### (b)

By definition,

$$\Gamma(1) = \int_0^\infty e^{-x} x^0 dx = \int_0^\infty e^{-x} dx.$$

Thus

$$\Gamma(1) = \left[-e^{-x}\right]_0^\infty = 0 - (-1) = 1.$$

### (c)

Let $p = n$ be a positive integer. From part (a),

$$\Gamma(n+1) = n\Gamma(n).$$

Repeatedly applying this formula gives

$$\Gamma(n+1) = n(n-1)(n-2)\cdots 1 \cdot \Gamma(1).$$

Since $\Gamma(1) = 1$,

$$\Gamma(n+1) = n(n-1)(n-2)\cdots 1 = n!.$$

### (d)

Using the recurrence formula repeatedly,

$$\Gamma(p+n) = (p+n-1)\Gamma(p+n-1).$$

Then

$$\Gamma(p+n) = (p+n-1)(p+n-2)\Gamma(p+n-2).$$

Continuing this process,

$$\Gamma(p+n) = (p+n-1)(p+n-2)\cdots(p+1)p\Gamma(p).$$

Thus

$$\boxed{p(p+1)(p+2)\cdots(p+n-1) = \frac{\Gamma(p+n)}{\Gamma(p)}}.$$

Now use $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

First,

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}.$$

Next,

$$\Gamma\left(\frac{11}{2}\right) = \frac{9}{2}\Gamma\left(\frac{9}{2}\right) = \frac{9}{2}\cdot\frac{7}{2}\Gamma\left(\frac{7}{2}\right) = \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\Gamma\left(\frac{5}{2}\right).$$

Continue until $\Gamma\left(\frac{1}{2}\right)$:

$$\Gamma\left(\frac{11}{2}\right) = \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right).$$

Therefore,

$$\Gamma\left(\frac{11}{2}\right) = \frac{9\cdot 7\cdot 5\cdot 3\cdot 1}{2^5}\sqrt{\pi} = \frac{945}{32}\sqrt{\pi}.$$

# 2

## (a)

For $p > -1$ and $s > 0$,

$$\mathcal{L}t^p \int_0^\infty e^{-st}t^p dt.$$

Let

$$x = st, \quad t = \frac{x}{s}, \quad dt = \frac{dx}{s}.$$

Then

$$\begin{aligned}\mathcal{L}t^p &= \int_0^\infty e^{-x}\left(\frac{x}{s}\right)^p \frac{dx}{s} \\ &= \frac{1}{s^{p+1}}\int_0^\infty e^{-x}x^p dx \\ &= \frac{\Gamma(p+1)}{s^{p+1}}, \quad s > 0.\end{aligned}$$

Therefore,

$$\boxed{\mathcal{L}t^p = \frac{\Gamma(p+1)}{s^{p+1}}, \quad p > -1,\ s > 0.}$$

## (b)

If $p = n$ is a positive integer, then from Problem 1,

$$\Gamma(n+1) = n!.$$

Substituting $p = n$ into part (a),

$$\mathcal{L}t^n = \frac{\Gamma(n+1)}{s^{n+1}} = \frac{n!}{s^{n+1}}.$$

## (c)

Take $p = -\frac{1}{2}$ in part (a). Since $-\frac{1}{2} > -1$,

$$\mathcal{L}t^{-1/2} \int_0^\infty e^{-st}t^{-1/2} dt.$$

Let

$$x = \sqrt{st}, \quad t = \frac{x^2}{s}, \quad dt = \frac{2x}{s}dx.$$

Also,

$$t^{-1/2} = \left(\frac{x^2}{s}\right)^{-1/2} = \frac{\sqrt{s}}{x}.$$

Thus,

$$\begin{aligned}\mathcal{L}t^{-1/2} &= \int_0^\infty e^{-x^2}\frac{\sqrt{s}}{x}\frac{2x}{s}dx \\ &= \frac{2}{\sqrt{s}}\int_0^\infty e^{-x^2}dx.\end{aligned}$$

So,

$$\boxed{\mathcal{L}t^{-1/2}\frac{2}{\sqrt{s}}\int_0^\infty e^{-x^2}dx, \quad s>0.}$$

Since

$$\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2},$$

we get

$$\boxed{\mathcal{L}t^{-1/2}\sqrt{\frac{\pi}{s}}, \quad s>0.}$$

## (d)

Take $p=\frac{1}{2}$ in part (a). Then

$$\mathcal{L}t^{1/2}\frac{\Gamma(3/2)}{s^{3/2}}.$$

Using the Gamma function recurrence from Problem 1,

$$\Gamma(p+1) = p\Gamma(p).$$

Let $p=\frac{1}{2}$. Then

$$\Gamma\left(\frac{3}{2}\right)\frac{1}{2}\Gamma\left(\frac{1}{2}\right).$$

From part (c),

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-x}x^{-1/2}dx = \sqrt{\pi}.$$

Therefore,

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

Hence,

$$\mathcal{L}t^{1/2} = \frac{\sqrt{\pi}/2}{s^{3/2}} = \boxed{\frac{\sqrt{\pi}}{2s^{3/2}}, \quad s>0.}$$

# 3

## (i)

Let $Y(s) = \mathcal{L}y(t)$. Taking Laplace transform,

$$\mathcal{L}y'' + 2y' + 5y = 0.$$

Using $y(0)=2,\ y'(0)=-1$,

$$(s^2Y - 2s + 1) + 2(sY - 2) + 5Y = (s^2+2s+5)Y - 2s - 3 = 0.$$

so

$$Y(s) = \frac{2s+3}{s^2+2s+5}Y(s) = \frac{2(s+1)}{(s+1)^2+4} + \frac{1}{(s+1)^2+4}.$$

Since

$$\mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+4}\right\} = e^{-t}\cos 2t,$$

and

$$\mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2+4}\right\} = e^{-t}\sin 2t,$$

we get

$$\boxed{y(t) = 2e^{-t}\cos 2t + \frac{1}{2}e^{-t}\sin 2t.}$$

## (ii)

Let $Y(s) = \mathcal{L}y(t)$. Taking Laplace transform,

$$\mathcal{L}y^{(4)} - 4y''' + 6y'' - 4y' + y = 0.$$

Given

$$y(0) = 0, \quad y'(0) = 1, \quad y''(0) = 0, \quad y'''(0) = 1,$$

we have

$$\mathcal{L}y^{(4)} = s^4Y - s^2 - 1,$$
$$\mathcal{L}y''' = s^3Y - s,$$
$$\mathcal{L}y'' = s^2Y - 1,$$
$$\mathcal{L}y' = sY.$$

Therefore,

$$(s^4Y - s^2 - 1) - 4(s^3Y - s) + 6(s^2Y - 1) - 4sY + Y = 0.$$

So

$$(s^4 - 4s^3 + 6s^2 - 4s + 1)Y - s^2 + 4s - 7 = 0.$$

Since

$$s^4 - 4s^3 + 6s^2 - 4s + 1 = (s-1)^4,$$

we obtain

$$Y(s) = \frac{s^2 - 4s + 7}{(s-1)^4}.$$

Rewrite the numerator in powers of $(s-1)$:

$$s^2 - 4s + 7 = (s-1)^2 - 2(s-1) + 4.$$

Thus

$$Y(s) = \frac{1}{(s-1)^2} - \frac{2}{(s-1)^3} + \frac{4}{(s-1)^4}.$$

Using

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-a)^{n+1}}\right\} = \frac{t^n}{n!}e^{at},$$

we get

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^2}\right\} = te^t,$$
$$\mathcal{L}^{-1}\left\{\frac{2}{(s-1)^3}\right\} = t^2e^t,$$
$$\mathcal{L}^{-1}\left\{\frac{4}{(s-1)^4}\right\} = \frac{2}{3}t^3e^t.$$

Therefore,

$$\boxed{y(t) = e^t\left(t - t^2 + \frac{2}{3}t^3\right).}$$

## 4

Using the definition

$$\mathcal{L}f(t)(s) = \int_0^\infty e^{-st}f(t)dt,$$

as in the Laplace transform notes and standard definition, we have

$$f(t) = 0 \quad \text{except on } [\pi, 2\pi).$$

Therefore,

$$\mathcal{L}f(t) = \int_\pi^{2\pi} e^{-st}(t-\pi)dt.$$

Let

$$u = t - \pi, \quad t = u + \pi, \quad dt = du.$$

When $t=\pi$, $u=0$; when $t=2\pi$, $u=\pi$. Thus,

$$\begin{aligned}\mathcal{L}f(t) &= \int_0^{\pi} e^{-s(u+\pi)}u\,du \\ &= e^{-\pi s}\int_0^{\pi} ue^{-su}\,du.\end{aligned}$$

By integration by parts,

$$\int_0^{\pi} ue^{-su}\,du = \left[-\frac{ue^{-su}}{s} - \frac{e^{-su}}{s^2}\right]_0^{\pi} \frac{1}{s^2} - e^{-\pi s}\left(\frac{\pi}{s}+\frac{1}{s^2}\right).$$

Hence,

$$\begin{aligned}\mathcal{L}f(t) &= e^{-\pi s}\left[\frac{1}{s^2} - e^{-\pi s}\left(\frac{\pi}{s}+\frac{1}{s^2}\right)\right] \\ &= \frac{e^{-\pi s}}{s^2} - \frac{\pi e^{-2\pi s}}{s} - \frac{e^{-2\pi s}}{s^2}.\end{aligned}$$

Therefore,

$$\boxed{\mathcal{L}f(t) = \frac{e^{-\pi s}-e^{-2\pi s}}{s^2} - \frac{\pi e^{-2\pi s}}{s}, \quad s>0.}$$

# 5

## (i)

Let $G(s)=\mathcal{L}g(t)$. Since

$$g(t) = \begin{cases}\sin t, & 0\le t<\pi,\\ 0, & t\ge\pi,\end{cases}$$

we can write

$$g(t) = \sin t + u_{\pi}(t)\sin(t-\pi).$$

Thus

$$G(s) = \frac{1}{s^2+1} + \frac{e^{-\pi s}}{s^2+1} = \frac{1+e^{-\pi s}}{s^2+1}.$$

Taking Laplace transforms of

$$y''+y'+\frac{5}{4}y = g(t), \qquad y(0)=0, \quad y'(0)=0,$$

gives

$$\left(s^2+s+\frac{5}{4}\right)Y(s) = G(s).$$

Hence

$$Y(s) = \frac{1+e^{-\pi s}}{(s^2+1)\left(s^2+s+\frac{5}{4}\right)}.$$

Now

$$\frac{1}{(s^2+1)\left(s^2+s+\frac{5}{4}\right)} = \frac{-\frac{16}{17}s+\frac{4}{17}}{s^2+1} + \frac{\frac{16}{17}s+\frac{12}{17}}{\left(s+\frac{1}{2}\right)^2+1}.$$

Therefore define

$$f(t) = -\frac{16}{17}\cos t + \frac{4}{17}\sin t + \frac{16}{17}e^{-t/2}\cos t + \frac{4}{17}e^{-t/2}\sin t.$$

By the shifting rule,

$$\boxed{y(t) = f(t) + u_{\pi}(t)f(t-\pi)}$$

where

$$\boxed{f(t) = -\frac{16}{17}\cos t + \frac{4}{17}\sin t + \frac{16}{17}e^{-t/2}\cos t + \frac{4}{17}e^{-t/2}\sin t.}$$

## (ii)

Taking Laplace transforms of

$$y^{(4)} - y = u_1(t) - u_2(t),$$

with

$$y(0) = y'(0) = y''(0) = y'''(0) = 0,$$

gives

$$(s^4 - 1)Y(s) = \frac{e^{-s}}{s} - \frac{e^{-2s}}{s}.$$

So

$$Y(s) = \left(e^{-s} - e^{-2s}\right)\frac{1}{s(s^4 - 1)}.$$

Now

$$\frac{1}{s(s^4 - 1)} = \frac{s}{2(s^2 + 1)} + \frac{1}{4(s + 1)} + \frac{1}{4(s - 1)} - \frac{1}{s}.$$

Thus

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^4 - 1)}\right\} = \frac{1}{2}\cos t + \frac{1}{4}e^{-t} + \frac{1}{4}e^{t} - 1.$$

Let

$$q(t) = \frac{1}{2}\cos t + \frac{1}{2}\cosh t - 1.$$

Then by the shifting rule,

$$\boxed{y(t) = u_1(t)q(t - 1) - u_2(t)q(t - 2)}$$

that is,

$$\boxed{y(t) = u_1(t)\left[\frac{1}{2}\cos(t - 1) + \frac{1}{2}\cosh(t - 1) - 1\right] - u_2(t)\left[\frac{1}{2}\cos(t - 2) + \frac{1}{2}\cosh(t - 2) - 1\right].}$$

# 6

## (i)

$$y'' - y = -20\delta(t - 3), \quad y(0) = 1, \quad y'(0) = 0.$$

Taking Laplace transforms,

$$(s^2Y - s) - (Y) = -20e^{-3s}.$$

Thus

$$(s^2 - 1)Y = s - 20e^{-3s},$$

so

$$Y = \frac{s}{s^2 - 1} - 20e^{-3s}\frac{1}{s^2 - 1}.$$

Since

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 - 1}\right\} = \cosh t, \quad \mathcal{L}^{-1}\left\{\frac{1}{s^2 - 1}\right\} = \sinh t,$$

the solution is

$$\boxed{y(t) = \cosh t - 20u_3(t)\sinh(t - 3).}$$

## (ii)

$$y'' + 2y' + 2y = \cos t + \delta(t - \pi/2), \quad y(0) = 0, \quad y'(0) = 0.$$

Taking Laplace transforms,

$$(s^2Y) + 2(sY) + 2Y = \frac{s}{s^2 + 1} + e^{-\pi s/2}.$$

Hence

$$Y = \frac{s}{(s^2 + 1)(s^2 + 2s + 2)} + \frac{e^{-\pi s/2}}{s^2 + 2s + 2}.$$

For the first part,

$$\frac{s}{(s^2 + 1)(s^2 + 2s + 2)} = \frac{\frac{1}{5}s + \frac{2}{5}}{s^2 + 1} + \frac{-\frac{1}{5}s - \frac{4}{5}}{(s + 1)^2 + 1}.$$

Rewrite

$$-\frac{1}{5}s - \frac{4}{5} = -\frac{1}{5}(s+1) - \frac{3}{5}.$$

Therefore,

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)(s^2+2s+2)}\right\} = \frac{1}{5}\cos t + \frac{2}{5}\sin t - \frac{1}{5}e^{-t}\cos t - \frac{3}{5}e^{-t}\sin t.$$

For the impulse part,

$$\frac{1}{s^2+2s+2} = \frac{1}{(s+1)^2+1} = \mathcal{L}e^{-t}\sin t.$$

So,

$$\mathcal{L}^{-1}\left\{e^{-\pi s/2}\frac{1}{(s+1)^2+1}\right\} = u_{\pi/2}(t)e^{-(t-\pi/2)}\sin(t-\pi/2).$$

Thus the solution is

$$\boxed{y(t) = \frac{1}{5}\cos t + \frac{2}{5}\sin t - \frac{1}{5}e^{-t}\cos t - \frac{3}{5}e^{-t}\sin t + u_{\pi/2}(t)e^{-(t-\pi/2)}\sin(t-\pi/2).}$$

# 7

Let

$$\Phi(s) = \mathcal{L}\{\phi(t)\}, \quad K(s) = \mathcal{L}\{k(t)\}, \quad F(s) = \mathcal{L}\{f(t)\}.$$

The integral term is a convolution:

$$\int_0^t k(t-\xi)\phi(\xi)\,d\xi = (k*\phi)(t).$$

By the convolution theorem,

$$\mathcal{L}\{(k*\phi)(t)\} = K(s)\Phi(s),$$

which is exactly the rule listed in the Laplace transform table of the uploaded notes:

$$\mathcal{L}\left\{\int_0^t f(t-\tau)g(\tau)d\tau\right\} = F(s)G(s).$$

Taking the Laplace transform of

$$\phi(t) + \int_0^t k(t-\xi)\phi(\xi)d\xi = f(t)$$

gives

$$\Phi(s) + K(s)\Phi(s) = F(s).$$

Thus

$$(1+K(s))\Phi(s) = F(s),$$

and therefore

$$\boxed{\mathcal{L}\{\phi(t)\} = \Phi(s) = \frac{F(s)}{1+K(s)} = \frac{\mathcal{L}\{f(t)\}}{1+\mathcal{L}\{k(t)\}}.}$$

Hence the solution of the original Volterra integral equation is

$$\boxed{\phi(t) = \mathcal{L}^{-1}\left\{\frac{\mathcal{L}\{f(t)\}}{1+\mathcal{L}\{k(t)\}}\right\}.}$$

# 8

## (a) Solve by Laplace transform

Here,

$$\int_0^t (t-\xi)\phi(\xi)d\xi$$

is the convolution of $k(t) = t$ and $\phi(t)$. Since

$$\mathcal{L}t = \frac{1}{s^2}, \qquad \mathcal{L}\sin 2t = \frac{2}{s^2+4},$$

taking Laplace transforms gives

$$\Phi(s) + \frac{1}{s^2}\Phi(s) = \frac{2}{s^2+4}.$$

Thus

$$\left(1 + \frac{1}{s^2}\right)\Phi(s) = \frac{2}{s^2+4}.$$

So

$$\Phi(s) = \frac{2}{s^2+4} \cdot \frac{s^2}{s^2+1} = \frac{2s^2}{(s^2+1)(s^2+4)}.$$

Use partial fractions:

$$\frac{2s^2}{(s^2+1)(s^2+4)} = \frac{A}{s^2+1} + \frac{B}{s^2+4}.$$

Then

$$2s^2 = A(s^2+4) + B(s^2+1),$$

so

$$A + B = 2, \qquad 4A + B = 0.$$

Hence

$$A = -\frac{2}{3}, \qquad B = \frac{8}{3}.$$

Therefore

$$\Phi(s) = -\frac{2}{3}\frac{1}{s^2+1} + \frac{8}{3}\frac{1}{s^2+4}.$$

Taking inverse Laplace transforms,

$$\boxed{\phi(t) = -\frac{2}{3}\sin t + \frac{4}{3}\sin 2t.}$$

## (b) Differentiate the integral equation twice

The equation is

$$\phi(t) + \int_0^t (t-\xi)\phi(\xi)d\xi = \sin 2t.$$

First put $t = 0$:

$$\phi(0) + 0 = \sin 0 = 0,$$

so

$$\boxed{\phi(0) = 0.}$$

Differentiate once:

$$\phi'(t) + \frac{d}{dt}\int_0^t (t-\xi)\phi(\xi)d\xi = 2\cos 2t.$$

By Leibniz rule,

$$\frac{d}{dt}\int_0^t (t-\xi)\phi(\xi)d\xi = \int_0^t \phi(\xi)d\xi.$$

Hence

$$\phi'(t) + \int_0^t \phi(\xi)d\xi = 2\cos 2t.$$

Put $t = 0$:

$$\phi'(0) + 0 = 2,$$

so

$$\boxed{\phi'(0) = 2.}$$

Differentiate once more:

$$\phi''(t) + \phi(t) = -4\sin 2t.$$

Thus

$$\boxed{\phi''(t) + \phi(t) = -4\sin 2t,}$$

with

$$\boxed{\phi(0) = 0, \qquad \phi'(0) = 2.}$$

## (c) Solve the IVP and verify

Solve

$$\phi'' + \phi = -4\sin 2t.$$

The complementary solution is

$$\phi_c(t) = c_1 \cos t + c_2 \sin t.$$

For a particular solution, try

$$\phi_p(t) = A\sin 2t.$$

Then

$$\phi_p'' + \phi_p = -4A\sin 2t + A\sin 2t = -3A\sin 2t.$$

Match with $-4\sin 2t$:

$$-3A = -4 \Rightarrow A = \frac{4}{3}.$$

So

$$\phi(t) = c_1 \cos t + c_2 \sin t + \frac{4}{3}\sin 2t.$$

Use $\phi(0) = 0$:

$$c_1 = 0.$$

Now

$$\phi'(t) = c_2 \cos t + \frac{8}{3}\cos 2t.$$

Use $\phi'(0) = 2$:

$$c_2 + \frac{8}{3} = 2,$$

so

$$c_2 = -\frac{2}{3}.$$

Therefore

$$\boxed{\phi(t) = -\frac{2}{3}\sin t + \frac{4}{3}\sin 2t.}$$

This is exactly the same solution as in part (a).